

Recurrent Memory Interference: Watching a Sequence Model Forget in Real Time

An interactive simulation exploring how fixed-capacity recurrent memory degrades as sequence length grows — built around a fast-weight synaptic-memory model.

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Introduction

Every recurrent model faces the same quiet crisis: it has to carry the past forward using a memory that never grows. A transformer can, in principle, look back at every token it has ever seen. A recurrent network cannot — it must compress everything it knows into a fixed-size state and keep updating that state, token after token, for as long as the sequence runs. When the amount of information trying to pass through that state exceeds what the state can hold, something has to give. Older traces blur into newer ones, distinct symbols start to resemble their neighbors, and the model's predictions quietly drift away from the ground truth. This phenomenon is called **recurrent memory interference (RMI)**.

Memory Lab is an interactive teaching tool built to make this abstract failure mode visible and tunable. Rather than describing interference in the abstract, it lets a learner directly control the variables that cause it — memory capacity, sequence length, and forgetting rate — and watch, step by step, how a model's internal state gets overwritten.

Core Concepts

1. Fixed-Capacity Memory vs. Sequence Length

A recurrent model's memory can be thought of as a small number of "slots," each capable of holding one activation pattern distinctly from the others. If a sequence is only as long as the number of available slots, every token can, roughly speaking, get its own place in memory. But as soon as the sequence outgrows the capacity, new tokens must be written into slots that already hold something else. The model is then forced to represent two or more pieces of information with overlapping activations — the definition of interference.

2. The Interference Score

Memory Lab tracks a single running number — the interference score — as a proxy for how "crowded" the memory has become at each step. It rises when capacity is exceeded and the model must borrow signal from neighboring tokens to represent a new one, and it eases when capacity comfortably covers

the sequence. This score is what turns an otherwise invisible internal process into something a learner can watch move in real time as the simulation steps forward.

3. Forgetting Rate

The forgetting rate controls how quickly older traces lose their signal strength on each step. A low forgetting rate preserves old information longer, which sounds beneficial, but it also means old traces linger and compete with new ones for the same limited slots — increasing interference. A high forgetting rate clears space quickly but at the cost of discarding information the model may still need. This is the classic stability–plasticity trade-off that shows up throughout memory research, from psychology to machine learning.

4. Fast-Weight Synaptic Memory

The simulation's underlying write rule is drawn from the family of "fast-weight" memory models, illustrated here through a Dragon-Hatchling-style synaptic update. At every step, the model writes a new key/value association into its weight matrix using an outer product, while the previous state decays by a factor lambda:

$$\mathbf{W}(\mathbf{t}) = \lambda \cdot \mathbf{W}(\mathbf{t}-1) + \mathbf{x}(\mathbf{t}) \cdot \mathbf{y}(\mathbf{t})^T$$

Here, $\mathbf{W}$ is the synaptic weight matrix acting as memory, $\mathbf{x}(\mathbf{t})$ and $\mathbf{y}(\mathbf{t})$ are the key and value associated with the current token, and lambda (0 to 1) is the decay term. Every new association is reinforced by the outer-product write; every prior association is simultaneously weakened by the multiplication with lambda. This single equation is the mechanical heart of the whole simulation: it is the reason capacity, sequence length, and forgetting rate all interact rather than acting independently.

The Project: What Memory Lab Simulates

Memory Lab presents the fast-weight update rule as a live, explorable system rather than a static equation. The interface is organized into five linked panels that together tell the story of a single sequence passing through the model:

- **Simulation controls.** Three sliders set the experiment: memory capacity (how many activations can stay distinct), sequence length (how many tokens pass through), and forgetting rate (how fast old traces decay).
- **Incoming signal.** A token-by-token view of the sequence stream, marking each token as actively stored, stored as a trace, or at risk of interference.
- **Active memory state.** A live map of activation intensity across memory slots, shown on a weak-to-strong scale, updated at every step.
- **Synaptic write panel.** A direct visualization of the fast-weight update itself, showing which connections are being reinforced, which are competing for the same slot, and which are overlapping.
- **Output check.** A side-by-side comparison of the ground-truth target sequence against what the degraded memory state actually predicts, making errors caused by interference directly visible.

The learner drives the whole thing: dragging a parameter, stepping forward one token at a time, or letting the simulation auto-play, watching the interference score and the memory heatmap respond immediately to each change.

Simulation Parameters at a Glance

Parameter	Range	Effect on Interference
Memory capacity	0 – 16 slots	Lower capacity relative to sequence length forces earlier and stronger interference.
Sequence length	0 – 32 tokens	Longer sequences increase the total pressure placed on a fixed memory.
Forgetting rate (lambda)	0% – 80%	Higher decay clears old traces faster, trading recall of older tokens for reduced crowding.

Observations and Results

Because Memory Lab is a hands-on, exploratory simulation rather than a benchmark suite, its "results" are best read as qualitative patterns that emerge consistently across runs, rather than as reported accuracy figures. Three patterns stand out.

Interference is negligible while capacity $\geq$ sequence length.

When there are at least as many memory slots as tokens, each token can occupy its own slot and the interference score stays close to zero for the whole run. The model output tracks the ground-truth target almost exactly.

Interference rises sharply once capacity is exceeded.

The moment sequence length overtakes capacity, the simulation shows new tokens "borrowing signal" from nearby stored traces — visible directly in the synaptic write panel as overlapping connections. The interference score climbs, and the output-check panel starts showing mismatches between predicted and expected tokens, concentrated on the oldest, most-overwritten entries first.

Forgetting rate reshapes, but doesn't remove, the trade-off.

Raising lambda's decay reduces crowding by clearing old traces faster, which lowers interference on later tokens but increases the chance that early-sequence information is lost outright before it's needed. Lowering the forgetting rate has the opposite effect: better recall of early tokens, but heavier overlap and higher interference once capacity is exceeded. No single setting eliminates the trade-off — the simulation makes clear that capacity, sequence length, and decay are three knobs on the same underlying pressure, not three independent variables.

Why This Matters

Recurrent memory interference isn't just a toy problem. It is a direct, mechanistic explanation for why classical RNNs and LSTMs struggle with long sequences, and why later architectures introduced gating,

larger hidden states, or attention over the full sequence history to work around it. Fast-weight and linear-attention models (of which the BDH-style update used here is one example) are an active area of research precisely because they offer a middle ground: constant-size recurrent state with an update rule that is carefully designed to minimize the kind of interference this simulation makes visible. Understanding the failure mode is a prerequisite for understanding why these newer designs matter.

Conclusion

Memory Lab turns a subtle, easy-to-hand-wave-away idea — that recurrent memory is finite and can be overloaded — into something a learner can see and manipulate directly. By exposing the fast-weight update rule, the live activation heatmap, and the resulting prediction errors side by side, the simulation connects a single line of mathematics (the decay-and-outer-product update) to its visible consequences one step at a time. The core lesson generalizes well beyond this one simulation: any system with a fixed-size memory and an unbounded input stream will eventually have to trade precision for capacity — and how gracefully it does so is, in large part, what separates one sequence model architecture from another.

Try it yourself: set memory capacity below sequence length in the live simulation and step forward to watch interference build in real time.